



BUILDING R13

Group 5

Group Members:

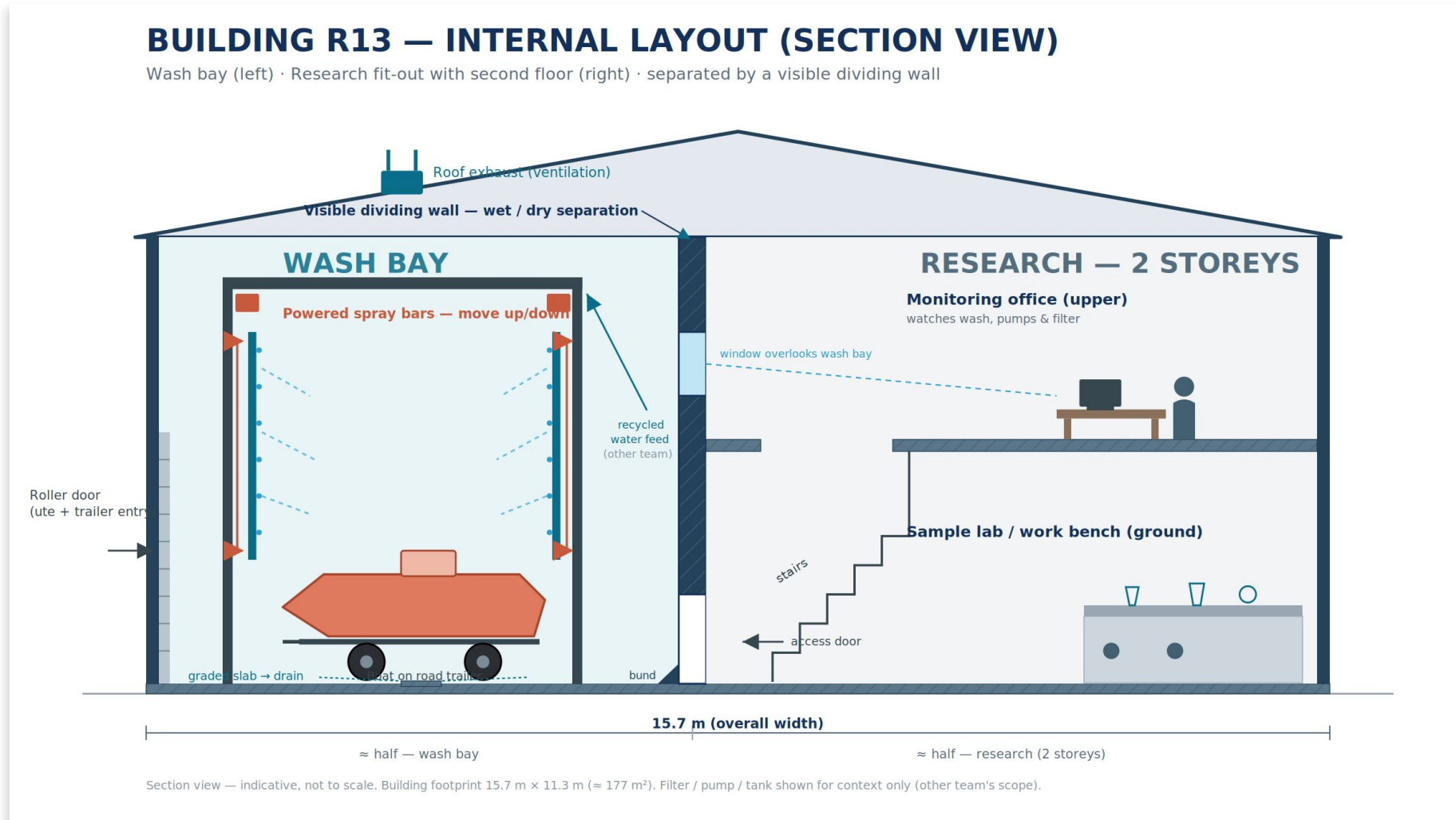
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Internal layout — section view



Wash bay on the left, two storey research fit out on the right, split by a dividing wall that keeps water off the dry side.

How we wash the boat

The chosen approach. I compared other options too (see backup).

Hold the boat

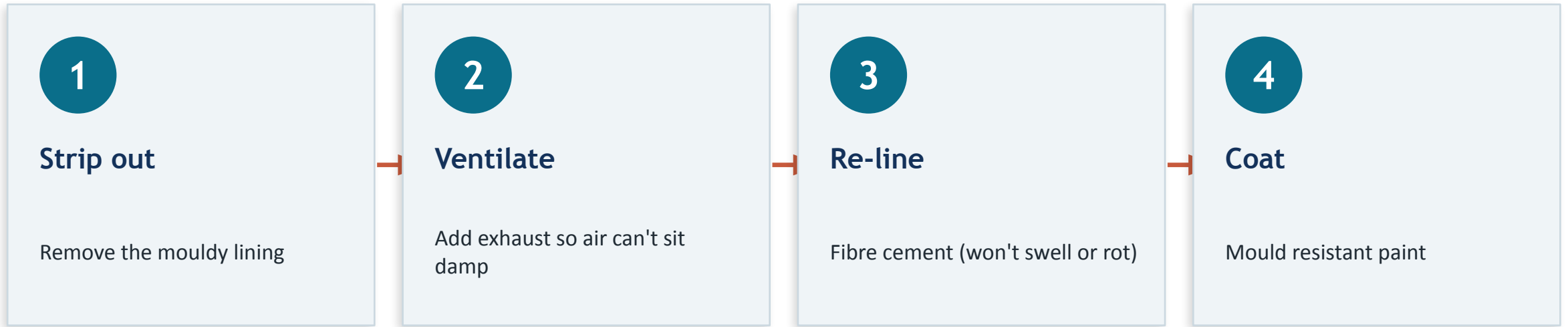
- Stays on its road trailer, towed in by the ute
- Cheapest, because we reuse gear we already have
- Safest, because nobody lifts a boat over people

Wash the boat

- Two spray bars move up and down to follow the hull
- A hand wand covers the tricky spots
- Touchless and water efficient, runs off recycled water

Walls — stop the mould coming back

Paint alone fails. Remove the mould, fix the damp, change the material.



Four layers of defence: remove the mould, the board it grew on, the damp air feeding it, and add a barrier on top. Indicative cost for the wet half is about \$16,000.

Cost and takeaways

≈ \$44,000

ex GST · range \$33k–\$55k · my part only

- Boat stays on its trailer (safest, reuses the ute)
- Two spray bars do the washing, touchless
- Walls rebuilt properly so the mould stays gone
- Building split into a wash side and a research side

Realistic. Durable. Water efficient. Reuses what is already on site.



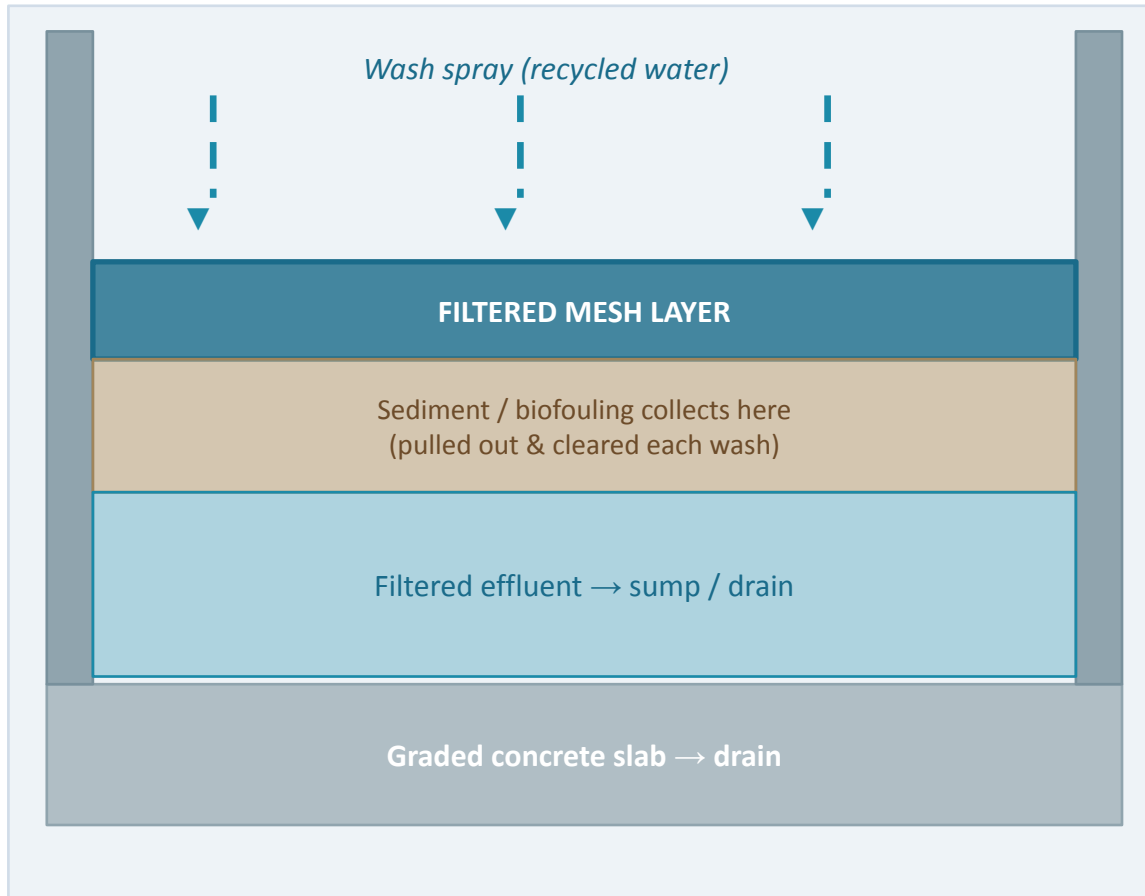
Amandhi's PART

Filtering and Drainage

Smart sensors and a camera that check the water, watch the pumps, and wash only where the boat is dirty.

Filtered mesh — keeping wash water off the drain

A passive filter layer that intercepts wash runoff before it reaches the drainage system.



1

What it catches

Captures biofouling, algae, silt, and hull debris from wash runoff before it reaches the drainage network or recycled water feed.

2

Where it sits

Recessed into the slab beneath the bunded wash bay floor. Sloped slab directs all runoff across the mesh before reaching the drain.

3

How it's maintained

Pull-out tray or lift-out frame. Inspected and cleared each wash cycle. Mesh replaced on a scheduled basis.

Mesh spec, cost & risks

Mesh type, frame, sizing, indicative cost, and how we manage the main risks.

Item	Specification	Cost (AUD)
Mesh material	316 stainless steel woven mesh — marine grade, salt-resistant	\$80–\$150
Mesh aperture	0.5–1 mm openings: fine enough for biofouling, fast-draining	—
Frame / tray	Galvanised steel pull-out frame, recessed into slab rebate	\$120–\$200
Spare meshes	2 replacements on site; rotate on 3-month inspection cycle	\$80–\$150
Slab rebate	Formed during concrete pour — no extra cost if designed in at build stage	\$0 if planned

Total indicative cost: **≈ \$280–\$500 AUD** · Fully passive — no power, no moving parts.

Risks & mitigations

Mesh blocks mid-wash

Heavy fouling could back-flood the bay. Mitigated by post-wash inspection and a secondary overflow path to drain.

Corrosion in salty air

316 SS resists salt spray well, but inspect for pitting at each swap. Replace on a 12-month maximum cycle regardless of condition.

Disposal of captured solids

Biofouling may contain invasive species — bag and bin as solid waste, never washed back to drain or sea.



FELIX'S PART

Monitor & ML

Smart sensors and a camera that check the water, watch the pumps, and wash only where the boat is dirty.

Monitoring & ML — what it does

The smart layer that keeps the wash efficient and the water clean.

1

Watch the water

Sensors track pH, turbidity, chlorine and TDS, so only clean water gets reused.

2

Watch the pumps

Sensors track pressure, flow and energy use, to catch a failing pump early.

3

Find the dirt

A camera and computer vision spot the stains, so the pumps target the dirty areas.

Main cost, main risk

Most of this part's cost is just hardware, the sensors and the camera.

The biggest risk is that sensors wear out in the salty air and give wrong readings, which could make the AI act on bad data. We handle it with routine inspections, planned replacements, and spares kept on site.

The hardware

Three pieces of kit. Most of this part's cost sits here.

What it watches	Example unit	Cost (AUD)	Check / replace
Water quality (pH, turbidity, chlorine, TDS)	Hanna HI98194 probe	\$2,300–3,000	1–2 months; electrode yearly, unit 2–3 yrs
Pumps (pressure, flow, energy use)	IFM SF5700 flow + pressure sensor	~\$970	Every 3 months; replace 3–4 yrs
Camera (finds stains on the hull)	LUCID Triton (IP67 camera)	\$480–1,200	Monthly; replace 3–5 yrs

Hardware total ≈ \$3,750–\$5,170 AUD the main spend for this part of the project

Finding the dirt — the smart bit

A camera plus computer vision so we only spray where it's needed.

How it works

- A self-trained YOLO model spots the stains and splits the hull into sections.
- The pumps focus water on the dirtiest sections, which saves water.
- The water and pump decisions are simple rules (thresholds and comparisons).
- The high-level "orchestrator" that ties it all together still needs more research.

Honest limits, and how we handle them

- The lens can fog or catch water mid-wash, and glare off a wet hull can confuse the model.
- AI models are about 99% accurate, so roughly 1% of reads can be off.
- Handled with a yearly hydrophobic lens coating, cross-checking the sensor readings, and regular inspections.

YEAMIN'S PART

SUSTAINABILITY

Built green from the ground up.

Water. Energy. Longevity. Every decision was made with the environment in mind.



Three pillars of sustainability

How R13 is designed to minimise its footprint at every stage.



Water efficiency

Recycled water powers every wash. Sensors confirm only clean water re-enters the system. Touchless spray bars target only the dirty sections, cutting total water use significantly compared to a manual hose-down.



Waste reduction

The boat stays on its existing trailer — no new lifting equipment needed. Wall materials were chosen to last decades without replacement, reducing the waste cycle that comes with repeated mould-driven strip-outs.



Durable materials

Fibre cement lining won't swell, rot, or grow mould. Unlike the original sheeting it replaces, it won't need pulling out and landfilling in two years. One good fix versus repeated short fixes.

Water recycling system

100%

of wash water
is recycled

*Closed-loop system. Water only leaves via evaporation
or controlled discharge when sensors detect it has
exceeded quality thresholds.*

How the loop works

1

Wash runs

Recycled water feeds the spray bars. Powered bars track the hull; only stained sections receive full pressure.

2

Water drains to bund

The graded slab channels all runoff into the bunded drain. Nothing escapes to stormwater.

3

Sensors check quality

pH, turbidity, chlorine, and TDS are measured in real time. Clean water is cleared for reuse.

4

Back to tank

Cleared water re-enters the supply tank. Only water that fails quality thresholds is held for disposal.

Smart monitoring — less water, less waste

The AI layer does more than save time — it is the environmental control system.



Targeted spray

Computer vision identifies dirty hull sections. Pumps only activate where stains are detected — no blanket wash.



Water quality gate

pH, turbidity & TDS sensors act as a gate: contaminated water cannot re-enter the system or escape to drain.



Pump health alerts

Pressure and flow monitoring catches faults early, preventing leaks or uncontrolled discharges before they happen.



Closed-loop proof

Logged sensor data provides an auditable record of water quality and reuse volumes for environmental reporting.

Result: every litre is tracked, every wash is targeted, and every discharge decision is data-driven.

Sustainability summary

- Recycled water — closed-loop wash, nothing to stormwater
- Targeted spray — AI cuts water use vs blanket hosing
- Sensor gating — contaminated water never re-enters supply
- Fibre cement walls — decades of service, no repeat strip-outs
- Reuse existing ute & trailer — no new equipment made
- Auditable logs — sensor data supports env. reporting

100%

wash water recycled

~1%

AI error rate — cross-checked

2–3×

wall lifespan vs original lining

0

new vehicles or lifts needed

Realistic. Durable. Water efficient. Reuses what is already on site.